

5. Write a program for Implementation of K-Nearest Neighbours (K-NN) in Python

Aim

To implement the **K-Nearest Neighbours (K-NN) algorithm** in Python for classification and predict the class of a new data point.

Algorithm

1. Import the required Python libraries.
2. Create a sample dataset with features and class labels.
3. Split the dataset into training and testing data.
4. Create a K-NN classifier with a suitable value of **K**.
5. Train the classifier using the training data.
6. Predict the classes of the test data.
7. Calculate the accuracy of the model.
8. Predict the class of a new data point.
9. Display the predicted result and accuracy.

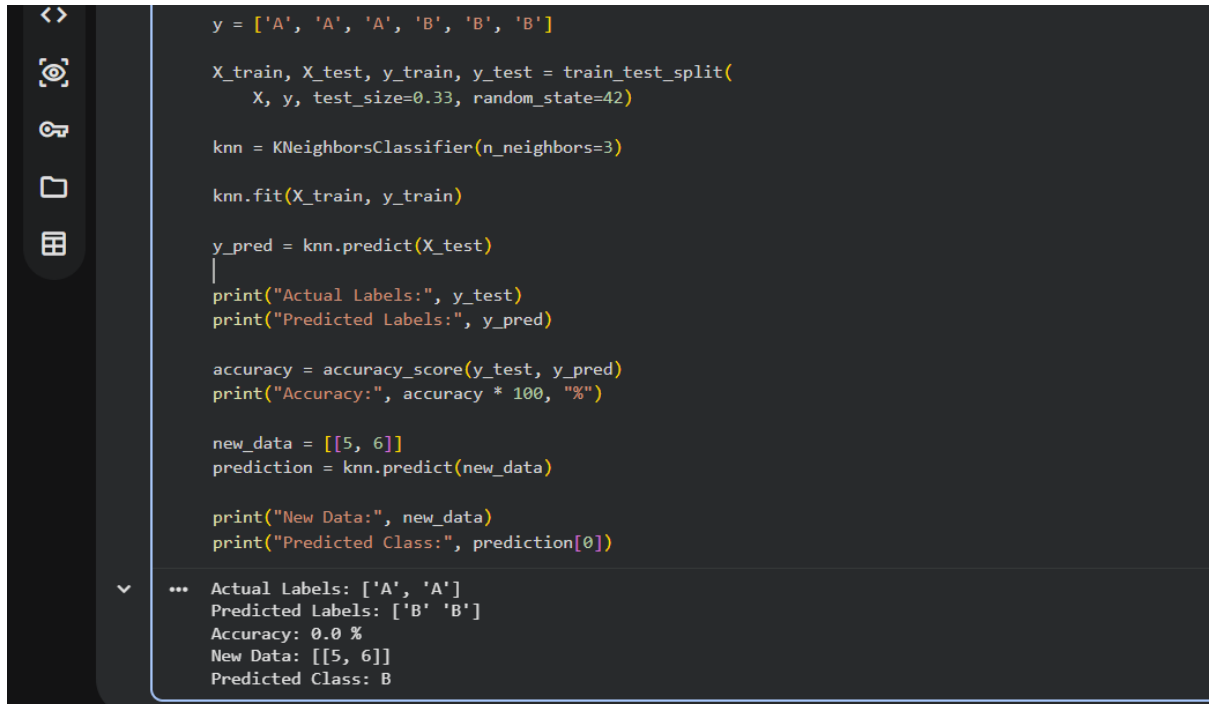
PROGRAM:

```
from sklearn.neighbors import KNeighborsClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score

X = [[1, 2], [2, 3], [3, 3],
      [6, 7], [7, 8], [8, 8]]
y = ['A', 'A', 'A', 'B', 'B', 'B']
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.33, random_state=42)
knn = KNeighborsClassifier(n_neighbors=3)
knn.fit(X_train, y_train)
y_pred = knn.predict(X_test)
print("Actual Labels:", y_test)
print("Predicted Labels:", y_pred)
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy * 100, "%")
new_data = [[5, 6]]
```

```
prediction = knn.predict(new_data)
print("New Data:", new_data)
print("Predicted Class:", prediction[0])
```

OUTPUT:



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y = ['A', 'A', 'A', 'B', 'B', 'B']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.33, random_state=42)

knn = KNeighborsClassifier(n_neighbors=3)

knn.fit(X_train, y_train)

y_pred = knn.predict(X_test)
|
print("Actual Labels:", y_test)
print("Predicted Labels:", y_pred)

accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy * 100, "%")

new_data = [[5, 6]]
prediction = knn.predict(new_data)

print("New Data:", new_data)
print("Predicted Class:", prediction[0])

▼ ... Actual Labels: ['A', 'A']
Predicted Labels: ['B' 'B']
Accuracy: 0.0 %
New Data: [[5, 6]]
Predicted Class: B
```

RESULT:

Thus, the **K-Nearest Neighbours (K-NN)** algorithm was successfully implemented in Python and the class of a new data point was predicted successfully.